

BUILT — Schematic legibility viewer (v0.27.0)

Same drawing, made readable: a 'Clean' toggle + contrast (server-side, PyMuPDF→Pillow), a 3D tilt of the flat page, and a hover loupe. Nothing invented; cited to its page.

VIEWER TOOLBAR (new controls)



REQUEST FLOW (offline, same dataset)



INVARIANTS & GROUNDING (R1 · R6)

Pure enhancement of the real page — contrast, sharpen, de-speckle, optional high-contrast/binarize — plus a flat-sheet 3D tilt and a magnifier. No strokes, depth, or detail are fabricated. Off by default: the dataset, FTS search, and the 104th sheet are untouched; toggles are presentation-only and reversible. Cleanup runs server-side via Pillow (auto-installed by run_app.bat); loupe + tilt are client-side CSS.

WHAT'S NEXT (proposed, not yet built)

- **Deep-zoom tiles (B)** OpenSeadragon tile pyramid for buttery zoom/pan to any portion — larger build.
- **Spotlight / box-highlight (C)** drag a region to dim everything outside it — pairs with the loupe.
- **Callout hotspots → parts (D)** hover a FIG number to highlight + open the NSN — OCR-box gated, grows as OCR completes.
- **Vectorize line art (E)** trace clean schematics to SVG for infinite-zoom crispness — optional, high effort.